Mahmoud Alyosify

Software Engineer | .NET & Python Backend, Data Systems & AI-Enabled Applications
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PROFESSIONAL SUMMARY

Software Engineer with production experience on **national-scale infrastructure** — backend systems for Egyptian toll operations serving **50%+ of national vehicles** — built in **C#, .NET, Entity Framework and SQL Server**. Currently an **AI Engineer Intern at Injaz Tech**, building a tested **FastAPI** service (136 automated tests, GitHub Actions CI, Docker) around an LLM application. Full-stack trained (**Route Academy .NET & Angular diploma**, plus **300+ hours** of ITI development courses in .NET, SQL Server, MEARN stack and Android), and equally comfortable shipping **Python services with FastAPI, Docker and AWS**. Now completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship** — bringing the increasingly rare combination of **solid software engineering fundamentals** and **genuine AI/ML depth** to teams building AI-enabled products.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Cloud Computing, End-to-End ML Pipelines, Large Language Models, Deep Learning, Reinforcement Learning**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Foundations in data structures, algorithms, operating systems, databases and discrete mathematics; approx. **100 competitive-programming problems** solved on Codeforces.
- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Backend & .NET	C#, .NET, ASP.NET Core, ASP.NET MVC, Web API, Entity Framework, ADO.NET, LINQ, WinForms, Crystal Reports, Stored Procedures, Triggers, RESTful API Design
Python & Services	Python, FastAPI, Pydantic, Flask, Streamlit, pytest, Docker, Docker Compose, AWS, CI/CD (GitHub Actions), SSE Streaming, Rate Limiting, ONNX, Background Jobs & Pipelines
Frontend	Angular, TypeScript, JavaScript, HTML5, CSS, Bootstrap, React (MEARN training)
Databases	SQL Server, MySQL, MongoDB, Database Design & Normalisation, Query Optimisation, ETL Pipelines
AI Integration	LLM APIs, Prompt Engineering, RAG, LangChain, LangGraph, Fine-Tuning (LoRA, QLoRA), PyTorch, TensorFlow, Scikit-learn, FAISS, ChromaDB, Agentic Coding Tools (Claude Code, Cursor, Copilot)
Languages & Tools	C#, Python, C++, C, JavaScript, TypeScript, SQL, Go, Java, R, MATLAB, Git/GitHub, Linux, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built a **FastAPI** service around an LLM/RAG assistant: typed Pydantic request/response contracts, SSE streaming, per-session memory, a single structured JSON error format, liveness/readiness probes and a bounded worker pool for a synchronous model pipeline.
- Wrote a **136-test pytest** suite (offline fakes for the model and index, plus live tests) including regression tests for wiring bugs, with **ruff** linting, a **GitHub Actions** CI pipeline and a multi-stage **Docker** / docker-compose setup.
- Added rate limiting (slowapi), retries with backoff, a circuit breaker and a context-aware answer cache that cut repeat-question latency from **34.7 s to 1.6 s**.

Machine Learning Instructor — *Aug 2026 – Sep 2026*
National Telecommunication Institute (NTI) — Arish & Assiut, Egypt

- Rehired to teach two **90-hour** applied ML bootcamps (**50 trainees**) at the Arish (online) and Assiut branches — regression, SVM, KNN, Naive Bayes, PCA and neural networks with Python labs, final exams and capstone-project evaluation.

Machine Learning Instructor — *Jul 2025 – Sep 2025*
National Telecommunication Institute (NTI) — Cairo, Egypt

- Delivered a **90-hour** hands-on ML program for engineering and CS students, guiding them through building complete solutions on real datasets.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported and maintained backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, working within the Control and Command environment.
- Co-developed an internal multi-department **data-management and reporting desktop application** using **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**, replacing manual reporting across stores and departments.
- Debugged and optimised **real-time transportation software** and provided technical support for legacy applications, minimising downtime and ensuring operational continuity.

Online Instructor — Programming, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Authored and taught Arabic-language courses in discrete mathematics, algorithms, probability, statistics and graph data mining for **10,000+ learners** (**700K+ views**).

SELECTED SOFTWARE PROJECTS

Inventory Management System
C# · ASP.NET Core · SQL Server · Barcode Scanning

- Built a CRUD-based inventory system with barcode scanning and alert notifications, covering database schema design through user-facing workflow.

E-Commerce Web Application
Angular · Bootstrap · Stripe API · JWT Authentication

- Developed a frontend shopping experience with secure JWT-based login and **Stripe** payment integration.

Mini-FAT File System
C · Systems Programming · Operating Systems

- Implemented a FAT-style file system in C from first principles — allocation tables, directory structures and file operations.

HORUS Sentinel — Multi-Agent Intelligence Platform *2026*
Python · Multi-Agent Architecture · Knowledge Graphs · Self-Hosted LLM

- Architected a distributed system of specialised autonomous agents writing into a shared knowledge-graph state layer, with a self-hosted model generating evidence-backed reports.

Additional Projects

- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on **AWS** as an interactive service.
- **AudioShield** — CNN-based audio classifier served through a Streamlit application with automated PDF reporting.
- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining pipeline analysing code-review effort across large open-source repositories.
- **Also:** Student Affairs System · Vitalism Solution (rPPG) · SCRAP · SimCLR-Vision-SSL (ONNX/FAISS) · ATM Transaction Analysis.

CERTIFICATIONS & AWARDS

Full Stack Development Diploma — .NET & Angular (Route Academy)
.NET Web Development — ITI (120 hrs)
SQL Server & C# — ITI (90 hrs)
MEARN Stack Development — ITI (60 hrs)
AWS Academy — ML Foundations, ML for NLP, Data Engineering
Microsoft Certified: Data Analyst Associate
1st Place — Smart Cities Hackathon 2022
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.